

Set-A

 Inspiring Excellence	BRAC University Department of Computer Science and Engineering Semester: Fall 2025 Course Code: CSE460 Course Title: VLSI Design	Final Examination Full Marks: 15 x 2 + 10 = 40 Time: 1 hour 45 minutes Date: 10 January 2026
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Student ID:	Name:	Section:
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[Question 1, Question 2 and Question 3 are mandatory. Question 4 is a bonus question.
Return the question paper with the answer script.]

Question 1: [CO4]

Scenario 1:

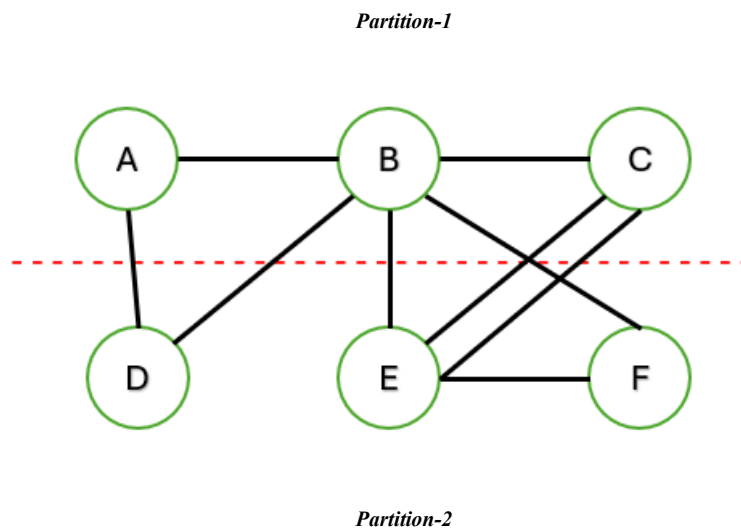


Fig.1

For question (a) and (b), an initial cut for the partition of 6 circuit elements, A, B, C, D, E, and F are given in **Fig.1** the cut lines divide the circuit elements into two partitions, **Partition-1** and **Partition-2**.

Scenario 2:

For question (c), you are given a chip layout where you have 2 distinct sources and 2 distinct targets. You have to reach **T1** from **S1** and **T2** from **S2**. The Lee Maze algorithm is an algorithm for signal routing, which uses a sequential approach.

Here, **X** is an obstacle for signal routing.

Question 2: [CO3]

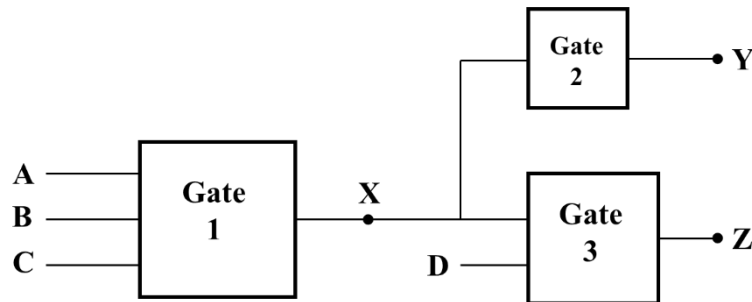


Fig. 2

The block diagram shown in **Fig. 2** represents a safety condition check system used in an industrial machine. Three safety signals are monitored as the inputs of the system:

- **A** = Door closed (1 = closed, 0 = open)
- **B** = Temperature within safe limit (1 = valid, 0 = invalid)
- **C** = Power supply ON (1 = ON, 0 = OFF)

When **all three safety conditions are satisfied** ($A=B=C=1$), **Gate 1** produces a **LOW** value at node **X**.

The signal at node X is used in two ways:

1. The low value at X is processed by **Gate 2** to generate an **enable signal** ($Y=1$) that allows the machine to operate.
2. X is combined with an independent **fault signal** (**D**, 1 = fault detected) to generate a **shutdown signal** (**Z**). A complete shutdown occurs ($Z=0$) when
 - A fault signal is present ($D=1$)
 - **or**, the safety conditions are violated ($X=1$)

The designer decided to implement the system with CMOS logic using a material where $\mu_n = 3\mu_p$.

(a)	Identify Gate 1, Gate 2 and Gate 3.	[3]
(b)	Draw the simplified RC circuit of Gate 1. [Choose widths of PMOS and NMOS transistors to achieve worst-case effective rise and fall resistance equal to that of a unit inverter (R)]	[4]
(c)	Determine the total capacitance at node X.	[5]
(d)	Derive the expression for propagation delay fall (t_{pdf}) and contamination delay rise (t_{cdr}) at node X.	[3]

Question 3: [CO3]

Suppose you are designing a chip using a **1V, 28 nm** technology that consists of **180 million transistors**, of which **30% are used to build logic gates**, while the remaining transistors are used in **memory arrays**. The average width of a logic transistor is 7λ , and the average width of a memory transistor is 4λ . The chip occupies an area of 50 mm^2

The gate capacitance and diffusion capacitance are $0.02 \text{ fF}/\mu\text{m}$ and $0.55 \text{ fF}/\mu\text{m}$, respectively, for the entire chip. Neglect wire capacitances. The operating **clock frequency** is **3.2 GHz**.

During every **100 clock cycles**, the logic transistors experience 12 switching events (On-Off pair), while the memory transistors experience 4 switching events (On-Off pair).

Total **short circuit power** consumption is **0.5W**.

You also obtain the following current data for each transistor:

- Gate leakage current = 3 nA/μm
- Junction leakage current = 1 nA/μm
- Subthreshold leakage current = 6 nA/μm

The acceptable total power consumption of the chip is 4 W.

(a)	Find the activity factor and switching frequency of the logic gates and the memory arrays .	[2]
(b)	Estimate the dynamic power and static power of the chip.	[6]
(c)	Calculate the total power consumption . Is the total power consumption within the acceptable range?	[2]

Question 4: [Bonus]

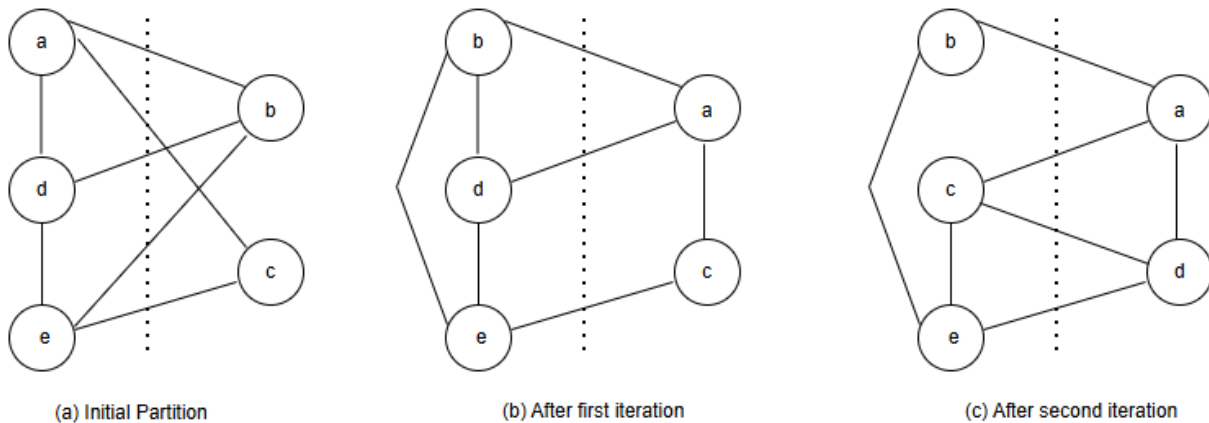


Fig. 3

(a)	From the scenario, identify if we need a 3rd iteration , if yes, perform the 3rd iteration , draw the new graph, and calculate the new cut cost.	[2]
(b)	You need to design a CMOS circuit to implement the logic function $Y = \overline{AB} + \overline{C}$ such that the worst case effective rise and fall resistance is equal to that of a unit inverter. Calculate the value of Kp and Kn for each transistor in this circuit. (Hint : The resistance of the two parallel paths must be the same).	[2]